

Apply filters to SQL queries

Project description

In this project, I used **SQL** to analyze authentication logs and employee records to support **cybersecurity monitoring and access control**. SQL queries were used to identify suspicious login behavior, detect unauthorized access attempts, and retrieve employee information based on departmental roles. This demonstrates how SQL plays a critical role in **threat detection, security auditing, and identity access management (IAM)**.

Retrieve after hours failed login attempts

Security teams monitor failed login attempts outside normal working hours to detect possible brute-force or unauthorized access attempts.

```
MariaDB [organization]> SELECT *  
->  
-> FROM log_in_attempts  
->  
-> WHERE login_time > '18:00' AND success = FALSE;  
-----+-----+-----+-----+-----+-----+-----+  
event_id | username | login_date | login_time | country | ip_address | success |  
-----+-----+-----+-----+-----+-----+-----+  
-----+
```

Explanation

This query retrieves all failed login attempts that occurred **outside standard office hours (9 AM – 5 PM)**, which may indicate suspicious activity.

Retrieve login attempts on specific dates

Investigating login activity on particular dates helps during **incident response** and forensic analysis.

```
MariaDB [organization]> SELECT *
->
-> FROM log_in_attempts
->
-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address |
+-----+-----+-----+-----+-----+-----+
| access |
```

Explanation

This query retrieves all login attempts that occurred on a **specific date**, allowing analysts to review activity during a known incident window.

Retrieve login attempts outside of Mexico

Monitoring logins from unexpected countries helps identify **potential account compromise or unauthorized access**.

```
MariaDB [organization]> SELECT *
->
-> FROM log_in_attempts
->
-> WHERE NOT country LIKE 'MEX%';
+-----+-----+-----+-----+-----+-----+
| event_id | username | login_date | login_time | country | ip_address |
+-----+-----+-----+-----+-----+-----+
| access |
```

Explanation

This query returns login attempts originating **outside Mexico**, helping detect suspicious geographic access.

Retrieve employees in Marketing

Access control policies often depend on department-based roles.

```
MariaDB [organization]> SELECT *
->
-> FROM employees
->
-> WHERE department = 'Marketing' AND office LIKE 'East%';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1052	a192b174c940	jdarosa	Marketing	East-195
1075	x573y883z772	fbautist	Marketing	East-267
1088	k865l965m233	rgosh	Marketing	East-157
1103	NULL	randerss	Marketing	East-460
1156	a184b775c707	dellery	Marketing	East-417
1163	h679i515j339	cwilliam	Marketing	East-216

Explanation

This query retrieves all employees who belong to the **Marketing department**, useful for role-based access reviews

Retrieve employees in Finance or Sales

Certain departments handle sensitive data and require stricter security controls.

```
MariaDB [organization]> SELECT *
->
-> FROM employees
->
-> WHERE department = 'Finance' OR department = 'Sales';
```

employee_id	device_id	username	department	office
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Explanation

This query retrieves employees from **Finance or Sales**, departments that typically handle confidential business information.

Retrieve all employees not IT

Identifying non-IT users helps enforce **least-privilege access**.

```
MariaDB [organization]> SELECT *  
->  
-> FROM employees  
->  
-> WHERE NOT department = 'Information Technology';  
+-----+-----+-----+-----+-----+  
| employee_id | device_id | username | department | office |  
+-----+-----+-----+-----+-----+
```

Explanation

This query lists all employees **excluding IT staff**, which is useful when auditing access rights or applying security restrictions.

Summary

This project demonstrates how **SQL is used in cybersecurity** to monitor login activity and manage employee access. SQL queries were applied to identify **failed after-hours login attempts**, analyze login activity on specific dates, and detect access from **outside authorized locations**. Additionally, employee records were queried to retrieve users by department, supporting **role-based access control** and **least-privilege principles**. Overall, this project highlights the importance of SQL in **threat detection, security auditing, and identity and access management (IAM)**.